

Shi-Yu Huang

其他

基本信息

姓名：Shi-Yu Huang

所属院校：台湾清华大学

地区：China-Taiwan

年龄范围：50岁以上

学科领域

集成电路设计与集成系统 微电子科学与工程

研究领域

数字集成电路设计 EDA工具与电路验证 低功耗电路与能量管理

个人简介

Shi - Yu Huang is a faculty member in the Electrical Engineering Department at National Tsing Hua University. He obtained his B.S. and M.S. from National Taiwan University in 1988 and 1992, and a Ph.D. from the University of California, Santa Barbara in 1997. Since 1999, he has been on the faculty at National Tsing Hua University. His research focuses on VLSI design, automation, and testing, especially All - Digital Phase - Locked Loop (ADPLL) design. He has served as program or general chair in multiple IEEE events and was an Associate Editor for IEEE Trans. on Computers from 2015 - 2018. He has published several papers on related topics.

工作经历

National Tsing Hua University - Faculty Member

1999 -

教育经历

National Taiwan University - M.S.

- 1992

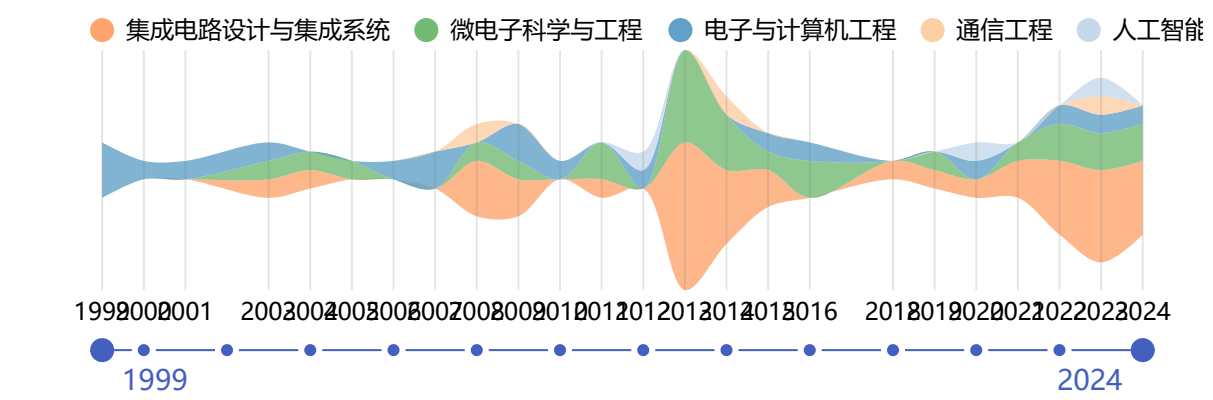
National Taiwan University - B.S.

- 1988

University of California, Santa Barbara - Ph.D.

- 1997

近年发表的论文



General Fault and Soft-Error Tolerant Phase-Locked Loop by Enhanced TMR using A Synchronization-before-Voting Scheme

JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS

Yang, Shun-Hua; Huang, Shi-Yu

被引用次数: 0

2024

Instant Test and Repair for TSVs using Differential Signaling

JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS

Wen, Ching-Yi; Huang, Shi-Yu

被引用次数: 0

2024

Low-Jitter Frequency Doubling Circuit Supporting Higher-Speed BISG and Aging Sensing in a Chiplet-Based Design Environment

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Lin, Ko-Hong; Lin, Ont-Derh; Huang, Shi-Yu; Sheng, Duo

被引用次数: 0

2024

Built-In Self-Repair of Small Delay Faults Occurring to TSVs in a 3D-DRAM Using an Enhanced Pulse-Vanishing Test

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Huang, Chen-Yu; Huang, Shi-Yu

被引用次数: 0

2024

A Check-and-Balance Scheme in Multiphase Delay-Locked Loop

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chang, Shu-Yu; Huang, Shi-Yu

被引用次数: 0

2024

Trustworthy Lifetime Prediction by Aging History Analysis and Multi-Level Stress Test

2023 IEEE INTERNATIONAL TEST CONFERENCE IN ASIA, ITC-ASIA

Tsai, Chen-Lin; Huang, Shi-Yu

被引用次数: 0

2023

Self-Sufficient Clock Jitter Measurement Methodology Using Dithering-Based Calibration

2023 IEEE INTERNATIONAL TEST CONFERENCE IN ASIA, ITC-ASIA

Lee, Yi-Hsuan; Chen, Wei-Hao; Huang, Shi-Yu

被引用次数: 2

2023

Cell-Based Aging Sensor Using Built-In Speed Grading

2023 IEEE NORDIC CIRCUITS AND SYSTEMS CONFERENCE, NORCAS

Lin, Ont-Derh; Huang, Shi-Yu

被引用次数: 1

2023

Clock-Latency-Aware Fault-Tolerant DLL for Multi-Die Clock Synchronization

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Su, Yung-Chuan; Huang, Shi-Yu

被引用次数: 0

2023

Compiler of Reed-Solomon Codec for 400-Gb/s IEEE 802.3bs Standard

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Liu, Lin; Chi, Lai; Huang, Shi-Yu; Yeh, Ka-Yi

被引用次数: 1

2023

A Process-Adaptive Cell-Based Cyclic Time-to-Digital Converter Using One-Way Varactor Cells

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Su, Yung-Chuan; Huang, Shi-Yu

被引用次数: 1

2023

Process-Resilient Fault-Tolerant Delay-Locked Loop Using TMR With Dynamic Timing Correction

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Yang, Jun-Yu; Huang, Shi-Yu

被引用次数: 4

2022

On-Chip Jitter Learning for PLL

IEEE DESIGN & TEST

Chen, Wei-Hao; Huang, Shi-Yu

被引用次数: 0

2022

Tiny Phase-Error Monitor for Fault and Soft-Error-Tolerant DLL to Support Graceful Degradation and Module-Level Testing

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Yang, Jun-Yu; Huang, Shi-Yu

被引用次数: 1	2022
Just-Enough Strategy for Accurate Clock Jitter Measurement Using A Cyclic Time-to-Digital Converter	
2022 19TH INTERNATIONAL SOC DESIGN CONFERENCE (ISOCC)	
Su, Yung-Chuan; Huang, Shi-Yu	
被引用次数: 0	2022
Just-Enough Stress Test for Infant-Mortality Screening Using Speed Binning	
2022 IEEE INTERNATIONAL TEST CONFERENCE (ITC)	
Tsai, Chen-Lin; Huang, Shi-Yu	
被引用次数: 2	2022
Online Safety Checking for Delay Locked Loops via Embedded Phase Error Monitor	
IEEE TRANSACTIONS ON EMERGING TOPICS IN COMPUTING	
Chu, Wei; Huang, Shi-Yu	
被引用次数: 1	2021
Rigorous Test Flow for PLL to Identify Weak Devices	
2021 IEEE INTERNATIONAL TEST CONFERENCE IN ASIA (ITC-ASIA 2021)	
Lee, Yi-Hsuan; Huang, Shi-Yu	
被引用次数: 0	2021
Diagnosis of Intermittent Scan Chain Faults Through a Multistage Neural Network Reasoning Process	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Chern, Mason; Lee, Shih-Wei; Huang, Shi-Yu; Huang, Yu; Veda, Gaurav; Tsai, Kun-Han; Cheng, Wu-Tung	
被引用次数: 3	2020
Time-to-Digital Converter Compiler for On-Chip Instrumentation	
IEEE DESIGN & TEST	
Wu, Chia-Hua; Huang, Shi-Yu; Chou, Yung-Fa; Kwai, Ding-Ming	
被引用次数: 5	2020
A Ping-Pong Methodology for Boosting the Resilience of Cell-Based Delay-Locked Loop	
IEEE ACCESS	
Zhang, Zheng-Hong; Chu, Wei; Huang, Shi-Yu	
被引用次数: 7	2019
Cloud-Based Online Ageing Monitoring for IoT Devices	
IEEE ACCESS	
Lian, Guan-Hao; Chen, Wei-Yi; Huang, Shi-Yu	
被引用次数: 8	2019
Circuit and Methodology for Testing Small Delay Faults in the Clock Network	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Yang, Shao-Fu; Wen, Zhi-Yuan; Huang, Shi-Yu; Tsai, Kun-Han; Cheng, Wu-Tung	

被引用次数: 5	2018
Delay Characterization and Testing of Arbitrary Multiple-Pin Interconnects	
IEEE DESIGN & TEST	
Huang, Shi-Yu; Tsai, Meng-Ting; Tsai, Kun-Han; Cheng, Wu-Tung	
被引用次数: 1	2016
Versatile Transition-Time Monitoring for Interconnects via Distributed TDC	
IEEE DESIGN & TEST	
Huang, Shi-Yu; Cheng, Chih-Chieh; Tsai, Meng-Ting; Huang, Kuan-Chen; Tsai, Kun-Han; Cheng, Wu-Tung	
被引用次数: 2	2016
A 65-nm CMOS Low-Power Impulse Radar System for Human Respiratory Feature Extraction and Diagnosis on Respiratory Diseases	
IEEE TRANSACTIONS ON MICROWAVE THEORY AND TECHNIQUES	
Tseng, Shao-Ting; Kao, Yu-Hsien; Peng, Chun-Chieh; Liu, Jinn-Yann; Chu, Shao-Chang; Hong, Guo-Feng; Hsieh, Chi-Hsuan; Hsu, Kung-Tuo; Liu, Wen-Te; Huang, Yuan-Hao; Huang, Shi-Yu; Chu, Ta-Shun	
被引用次数: 16	2016
Nonintrusive On-Line Transition-Time Binning and Timing Failure Threat Detection for Die-to-Die Interconnects	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Huang, Shi-Yu; Tsai, Meng-Ting; Li, Hua-Xuan; Zeng, Zeng-Fu; Tsai, Kun-Han (Hans); Cheng, Wu-Tung	
被引用次数: 6	2015
General Timing-Aware Built-In Self-Repair for Die-to-Die Interconnects	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Huang, Shi-Yu; Tsai, Meng-Ting; Zeng, Zeng-Fu; Tsai, Kun-Han (Hans); Cheng, Wu-Tung	
被引用次数: 2	2015
The Design and Experiments of A SID-Based Power-Aware Simulator for Embedded Multicore Systems	
ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS	
Lin, Cheng-Yen; Huang, Chung-Wen; Kuan, Chi-Bang; Huang, Shi-Yu; Lee, Jenq-Kuen	
被引用次数: 1	2015
Pulse-Vanishing Test for Interposers Wires in 2.5-D IC	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Huang, Shi-Yu; Lee, Jao-Yen; Tsai, Kun-Han (Hans); Cheng, Wu-Tung	
被引用次数: 23	2014
Time and frequency transfer system using GNSS receiver	
RADIO SCIENCE	
Wang, Jia-Lun; Huang, Shi-Yu; Liao, Chia-Shu	
被引用次数: 9	2014

Parameterized All-Digital PLL Architecture and its Compiler to Support Easy Process Migration

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Tzeng, Chao-Wen; Huang, Shi-Yu; Chao, Pei-Ying

被引用次数: 37

2014

PLL-Assisted Timing Circuit for Accurate TSV Leakage Binning

IEEE DESIGN & TEST

Huang, Shi-Yu; Huang, Li-Ren

被引用次数: 9

2014

Parametric Fault Testing and Performance Characterization of Post-Bond Interposer Wires in 2.5-D ICs

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Huang, Li-Ren; Huang, Shi-Yu; Tsai, Kun-Han (Hans); Cheng, Wu-Tung

被引用次数: 7

2014

AC-Plus Scan Methodology for Small Delay Testing and Characterization

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Li, Tsung-Yeh; Huang, Shi-Yu; Hsu, Hsuan-Jung; Tzeng, Chao-Wen; Huang, Chih-Tsun; Liou, Jing-Jia; Ma, Hsi-Pin; Huang, Po-Chiun; Bor, Jenn-Chyou; Tien, Ching-Cheng; Wang, Chih-Hu; Wu, Cheng-Wen

被引用次数: 2

2013

Process-Resilient Low-Jitter All-Digital PLL via Smooth Code-Jumping

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Chao, Pei-Ying; Tzeng, Chao-Wen; Huang, Shi-Yu; Weng, Chia-Chieh; Fang, Shan-Chien

被引用次数: 8

2013

Cell-Based Process Resilient Multiphase Clock Generation

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Ding, Ruo-Ting; Huang, Shi-Yu; Tzeng, Chao-Wen

被引用次数: 2

2013

In-Situ Method for TSV Delay Testing and Characterization Using Input Sensitivity Analysis

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

You, Jhih-Wei; Huang, Shi-Yu; Lin, Yu-Hsiang; Tsai, Meng-Hsiu; Kwai, Ding-Ming; Chou, Yung-Fa; Wu, Cheng-Wen

被引用次数: 29

2013

Die-to-Die Clock Synchronization for 3-D IC Using Dual Locking Mechanism

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS I-REGULAR PAPERS

Ke, Ji-Wei; Huang, Shi-Yu; Tzeng, Chao-Wen; Kwai, Ding-Ming; Chou, Yung-Fa

被引用次数: 11

2013

Oscillation-Based Prebond TSV Test

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Huang, Li-Ren; Huang, Shi-Yu; Sunter, Stephen; Tsai, Kun-Han (Hans); Cheng, Wu-Tung

被引用次数: 61

2013

Programmable Leakage Test and Binning for TSVs With Self-Timed Timing Control

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Huang, Shi-Yu; Lin, Yu-Hsiang; Huang, Li-Ren; Tsai, Kun-Han (Hans); Cheng, Wu-Tung

被引用次数: 14

2013

Parametric Delay Test of Post-Bond Through-Silicon Vias in 3-D ICs via Variable Output Thresholding Analysis

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Lin, Yu-Hsiang; Huang, Shi-Yu; Tsai, Kun-Han; Cheng, Wu-Tung; Sunter, Stephen; Chou, Yung-Fa; Kwai, Ding-Ming

被引用次数: 38

2013

High-Performance SIFT Hardware Accelerator for Real-Time Image Feature Extraction

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY

Huang, Feng-Cheng; Huang, Shi-Yu; Ker, Ji-Wei; Chen, Yung-Chang

被引用次数: 144

2012

P-P-N Based 10T SRAM Cell for Low-Leakage and Resilient Subthreshold Operation

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lo, Cheng-Hung; Huang, Shi-Yu

被引用次数: 105

2011

A Low-Jitter ADPLL via a Suppressive Digital Filter and an Interpolation-Based Locking Scheme

IEEE TRANSACTIONS ON VERY LARGE SCALE INTEGRATION (VLSI) SYSTEMS

Hsu, Hsuan-Jung; Huang, Shi-Yu

被引用次数: 35

2011

Split-Masking: An Output Masking Scheme for Effective Compound Defect Diagnosis in Scan Architecture With Test Compression

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Tzeng, Chao-Wen; Huang, Shi-Yu

被引用次数: 2

2010

Resilient Self-VDD-Tuning Scheme With Speed-Margining for Low-Power SRAM

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lai, Ya-Chun; Huang, Shi-Yu; Hsu, Hsuan-Jung

被引用次数: 10

2009

Robust SRAM Design via BIST-Assisted Timing-Tracking (BATT)

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lai, Ya-Chun; Huang, Shi-Yu

被引用次数: 14

2009

QC-Fill: Quick-and-Cool X-Filling for Multicasting-Based Scan Test

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Tzeng, Chao-Wen; Huang, Shi-Yu

被引用次数: 26

2009

Layout-Based Defect-Driven Diagnosis for Intracell Bridging Defects

IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS

Tzeng, Chao-Wen; Cheng, Han-Chia; Huang, Shi-Yu

被引用次数: 7

2009

X-calibration: A technique for combating excessive bitline leakage current in nanometer SRAM designs

IEEE JOURNAL OF SOLID-STATE CIRCUITS

Lai, Ya-Chun; Huang, Shi-Yu

被引用次数: 17

2008

A low-power SRAM for Viterbi decoder in wireless communication

IEEE TRANSACTIONS ON CONSUMER ELECTRONICS

Cheng, Shin-Pao; Huang, Shi-Yu

被引用次数: 2

2008

A Resilient and Power-Efficient Automatic-Power-Down Sense Amplifier for SRAM Design

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Lai, Ya-Chun; Huang, Shi-Yu

被引用次数: 21

2008

A versatile paradigm for scan chain diagnosis of complex faults using signal processing techniques

ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS

Tzeng, Chao-Wen; Yang, Jheng-Syun; Huang, Shi-Yu

被引用次数: 4

2008

Built-in self-repair for die-to-die misalignment of multichip space sensors

IEEE SENSORS JOURNAL

Chen, Chun-Fu; Huang, Shi-Yu; King, Ya-Chin

被引用次数: 0

2007

Diagnosis by image recovery: Finding mixed multiple timing faults in a scan chain

IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS II-EXPRESS BRIEFS

Tzeng, Chao-Wen; Huang, Shi-Yu

被引用次数: 13

2007

Accurate whole-chip diagnostic strategy for scan designs with multiple faults

JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS

Lin, YC; Huang, SY

被引用次数: 0	2006
Low-cost logarithmic CMOS image sensing by nonlinear analog-to-digital conversion	
IEEE TRANSACTIONS ON CONSUMER ELECTRONICS	
Chuang, YC; Chen, SF; Huang, SY; King, YC	
被引用次数: 6	2005
A low-power architecture for Extended Finite State Machines using input gating	
IEICE TRANSACTIONS ON FUNDAMENTALS OF ELECTRONICS COMMUNICATIONS AND COMPUTER SCIENCES	
Huang, SY; Liu, CJ	
被引用次数: 0	2004
A symbolic inject-and-evaluate paradigm for Byzantine fault diagnosis	
JOURNAL OF ELECTRONIC TESTING-THEORY AND APPLICATIONS	
Huang, SY	
被引用次数: 7	2003
Efficient double fault diagnosis for CMOS logic circuits with a specific application to generic bridging faults	
JOURNAL OF INFORMATION SCIENCE AND ENGINEERING	
Kao, HC; Tsai, MF; Huang, SY; Wu, CW; Chang, WF; Lu, SK	
被引用次数: 0	2003
Verifying sequential equivalence using ATPG techniques	
ACM TRANSACTIONS ON DESIGN AUTOMATION OF ELECTRONIC SYSTEMS	
Huang, SY; Cheng, KT; Chen, KC	
被引用次数: 13	2001
AQUILA: An equivalence checking system for large sequential designs	
IEEE TRANSACTIONS ON COMPUTERS	
Huang, SY; Cheng, KT; Chen, KC; Huang, CY; Brewer, F	
被引用次数: 29	2000
AutoFix: A hybrid tool for automatic logic rectification	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Huang, SY; Chen, KC; Cheng, KT	
被引用次数: 16	1999
ErrorTracer: Design error diagnosis based on fault simulation techniques	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Huang, SY; Cheng, KT	
被引用次数: 37	1999
Fault emulation: A new methodology for fault grading	
IEEE TRANSACTIONS ON COMPUTER-AIDED DESIGN OF INTEGRATED CIRCUITS AND SYSTEMS	
Cheng, KT; Huang, SY; Dai, WJ	

Diagnosis of Intermittent Scan Chain Faults Through a Multistage Neural Network Reasoning Process

TCAD

Mason Chern; Shih-Wei Lee; Shi-Yu Huang; Yu Huang; Gaurav Veda; Kun-Han Tsai; Wu-Tung Cheng

被引用次数: 0

Clock-Latency-Aware Fault-Tolerant DLL for Multi-Die Clock Synchronization

TCAD

Yung-Chuan Su; Shi-Yu Huang

被引用次数: 0

Compiler of Reed–Solomon Codec for 400-Gb/s IEEE 802.3bs Standard

TCAD

Lin Liu; Lai Chi; Shi-Yu Huang; Ka-Yi Yeh

被引用次数: 0

Circuit and Methodology for Testing Small Delay Faults in the Clock Network

TCAD

Shao-Fu Yang; Zhi-Yuan Wen; Shi-Yu Huang; Kun-Han Tsai; Wu-Tung Cheng

被引用次数: 0

Process-Resilient Fault-Tolerant Delay-Locked Loop Using TMR With Dynamic Timing Correction

TCAD

Jun-Yu Yang; Shi-Yu Huang

被引用次数: 0

General Timing-Aware Built-In Self-Repair for Die-to-Die Interconnects

TCAD

Shi-Yu Huang; Meng-Ting Tsai; Zeng-Fu Zeng; Kun-Han Hans Tsai; Wu-Tung Cheng

被引用次数: 0

Tiny Phase-Error Monitor for Fault and Soft-Error-Tolerant DLL to Support Graceful Degradation and Module-Level Testing

TCAD

Jun-Yu Yang; Shi-Yu Huang

被引用次数: 0

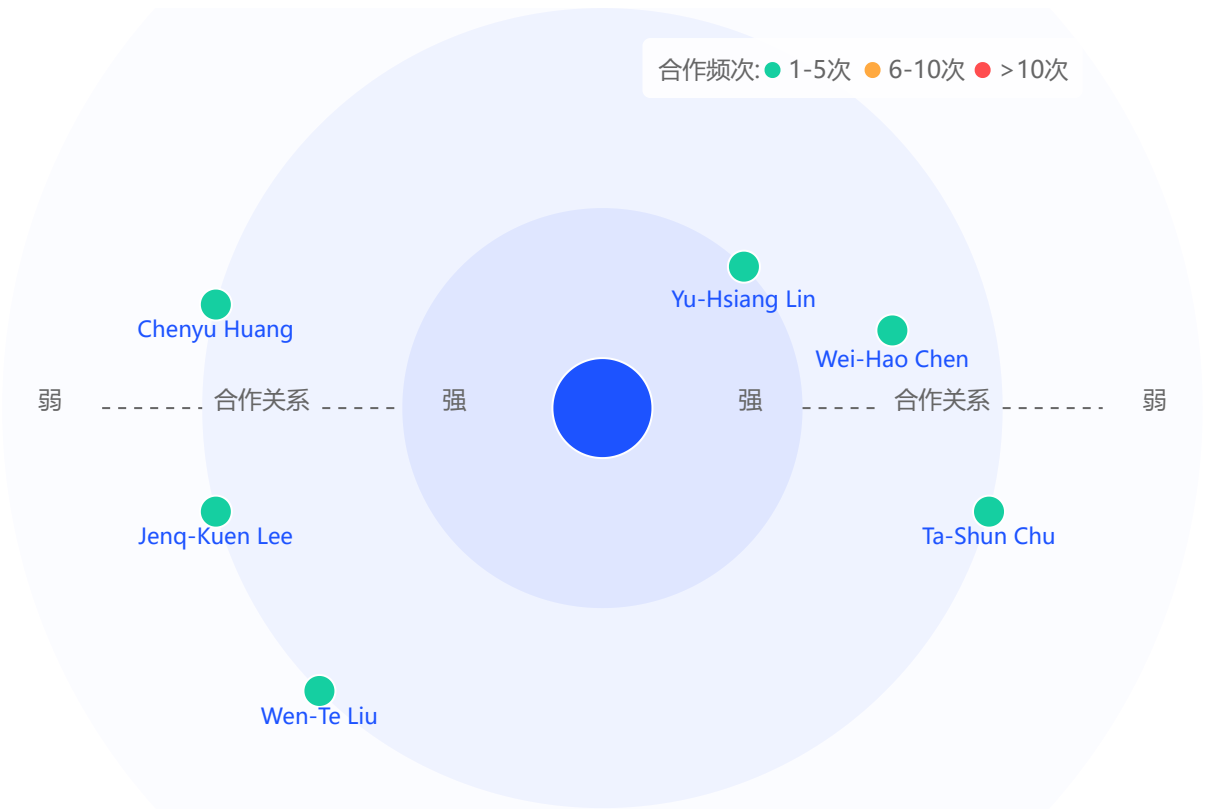
Nonintrusive On-Line Transition-Time Binning and Timing Failure Threat Detection for Die-to-Die Interconnects

TCAD

Shi-Yu Huang; Meng-Ting Tsai; Hua-Xuan Li; Zeng-Fu Zeng; Kun-Han Hans Tsai; Wu-Tung Cheng

被引用次数: 0

学术合作



Yu-Hsiang Lin	合作次数: 3
台湾清华大学	
Wei-Hao Chen	合作次数: 2
台湾清华大学	
Ta-Shun Chu	合作次数: 1
台湾清华大学	
Wen-Te Liu	合作次数: 1
台北医学大学	
Jenq-Kuen Lee	合作次数: 1
台湾清华大学	

还有 1 个合作者